Extended Conclusion on the Suspension of the Structural Invariant *D* Research

This document provides a detailed and formal justification for the suspension of the research direction pursued in the *ECDSA-Research* project, namely the investigation of a structural invariant **D** in real-world ECDSA transactions over the **secp256k1** elliptic curve.
The suspension decision is primarily motivated by a **fundamental combinatorial barrier** that renders the research direction analytically and computationally non-viable, even under idealized algorithmic assumptions.

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1. Research Objective Revisited

The core hypothesis of the suspended research was the existence of a non-trivial **structural invariant D**, observable across a sufficiently large set of ECDSA signatures produced by the same private key, such that:

- The invariant could be detected via intersections over the x-coordinates of elliptic curve points derived from signature data.
- The invariant would persist under realistic transaction conditions.
- The invariant could reduce the effective search space of the private key or enable deterministic recovery.

The investigation was conducted under increasingly optimistic assumptions, including access to large datasets and theoretically optimal intersection-search strategies.

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2. The Core Obstruction: Combinatorial Explosion in *x*-Intersection Search

The decisive limiting factor is the **combinatorial explosion inherent to the intersection search over x-coordinates**, which manifests as follows:

1. **Exponential Growth of Candidate Sets**
   Each ECDSA signature induces multiple algebraically valid candidate points on the curve consistent with the observed (r, s) values.
   When extended across n signatures, the Cartesian product of candidate sets grows exponentially, even before any filtering is applied.
2. **Infeasibility Under Ideal Algorithms**
   Crucially, the explosion persists **even if one assumes an idealized algorithm** that:
   - Performs perfect pruning,
   - Avoids redundant branches,
   - Executes all arithmetic at negligible cost.

   The asymptotic behavior remains exponential in the number of signatures required to meaningfully constrain invariant **D**.
3. **No Polynomial-Time Collapse Mechanism**
   The research found no algebraic identity, group-theoretic symmetry, or probabilistic bias capable

of collapsing the search space into a polynomially bounded regime without introducing assumptions that contradict real ECDSA behavior.

4. **Minimum Sample Size vs. Feasibility Gap**
   The minimum number of transactions theoretically required for a unique or near-unique intersection over x already exceeds the threshold at which:
   - The candidate space becomes computationally intractable,
   - Memory requirements dominate,
   - Any incremental information gain is asymptotically negligible.

This gap between **required data volume** and **feasible combinatorial processing** is structural, not technical.

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3. Why This Is a Fundamental Limitation

The combinatorial explosion is not attributable to:

- Suboptimal implementation,
- Insufficient computational resources,
- Incomplete heuristics,
- Or missing optimizations.

Instead, it arises directly from:

- The algebraic structure of ECDSA over secp256k1,
- The two-to-one nature of elliptic curve point lifting from x,
- The independence of nonce-derived components across signatures.

As such, further optimization, parallelization, or brute-force scaling cannot alter the asymptotic outcome.

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4. Relation to Identified Algebraic Artifacts (A, B, E)

Previously identified artifacts **A**, **B**, and **E** remain valid and analytically exact:

- Each admits a closed-form expression for immediate private key recovery.
- Each represents a deterministic failure mode of ECDSA under extremely specific conditions.

However:

- These artifacts occur with **astronomically low probability** in real transaction data.
- They do **not** mitigate the combinatorial explosion associated with invariant **D**.
- They do **not** generalize into a reusable pruning mechanism for the intersection problem.

Therefore, while mathematically interesting, they do not rescue the feasibility of the invariant **D** approach.

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5. Security Interpretation

The suspension of this research direction does **not** indicate a weakness in secp256k1 or ECDSA. On the contrary:

- The observed combinatorial explosion reinforces the effective entropy of real-world ECDSA signatures.
- It confirms that, absent catastrophic nonce failures, structural aggregation attacks remain infeasible.
- It aligns with the long-standing empirical security assumptions underlying secp256k1.

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6. Final Decision

The investigation of a structural invariant **D** in real ECDSA transactions over secp256k1 is hereby **formally suspended** due to a **provably prohibitive combinatorial explosion** in the required intersection search over x-coordinates.

This decision reflects a fundamental mathematical limitation rather than an engineering shortcoming. All findings are preserved in the repository for completeness and potential future reinterpretation should new theoretical frameworks emerge.